

Appendix 1. Port States in Each Processing Mode

Function	Pin function	Reset	Software Standby mode (SSTBY)		Deep Software Standby mode 1, 2, 3 (DSTBY1, 2, 3)		After Deep Software Standby mode is canceled (return to startup mode)	
			OPE = 0	OPE = 1	DSTBY1	DSTBY2/ DSTBY3	IOKEEP = 0	IOKEEP = 1 ¹
Mode	MD	Pull-up	Keep-I		Keep		Pull-up	Keep
JTAG/SWD	TCK/TMS/TDI/SWCLK	Pull-up	TCK/TDI/TMS/SWCLK input		TCK/TDI/TMS/SWCLK input		TCK/TDI/TMS/SWCLK input	
	TDO	Output	TDO output		TDO output		TDO output	
	SWDIO	Pull-up	SWDIO inout		SWDIO inout		SWDIO inout	
Trace	TCLK/TDATAx/SWO	TCLK/ TDATAx/SWO output	TCLK/TDATAx/SWO output		TCLK/TDATAx/SWO output		TCLK/TDATAx/SWO output	
IRQ	IRQx	Hi-Z	Hi-Z ²		Keep		Hi-Z	Keep
	IRQx-DS (x: Other than 5)	Hi-Z	Hi-Z ²		Keep ³		Hi-Z	Keep
	IRQ5-DS	Hi-Z	Hi-Z ²		Keep ³		Hi-Z	
AGT	AGTIO _n	Hi-Z	AGTIO _n inout		Keep		Hi-Z	Keep
	AGTO _n /AGTOA _n / AGTOB _n	Hi-Z	AGTO _n /AGTOA _n /AGTOB _n output		Keep		Hi-Z	Keep
ULPT	ULPTEEn/ULPTEVIn	Hi-Z	ULPTEEn/ULPTEVIn input		Keep		Hi-Z	Keep
	ULPTEEn-DS/ ULPTEVIn-DS	Hi-Z	ULPTEEn-DS/ULPTEVIn-DS input		ULPTEEn-DS/ ULPTEVIn-DS input	Hi-Z	Hi-Z	Keep
	ULPTOn/ ULPTOA _n / ULPTOB _n	Hi-Z	ULPTOn/ULPTOA _n /ULPTOB _n output		Keep		Hi-Z	Keep
	ULPTOn-DS/ ULPTOA _n - DS/ ULPTOB _n -DS	Hi-Z	ULPTOn/ULPTOA _n -DS/ ULPTOB _n -DS output		ULPTOn/ULPTOA _n - DS/ ULPTOB _n -DS output	Keep	Hi-Z	From DSTBY1: ULPTOn/ ULPTOA _n -DS/ ULPTOB _n -DS output From DSTBY1, 2: Keep
IIC	SCLn/SDAn	Hi-Z	Keep-O ²		Keep		Hi-Z	Keep
I3C	I3C_SCL0/I3C_SDA0	Hi-Z	Keep-O ²		Hi-Z		Hi-Z	
USBFS	USB_OVRCURx	Hi-Z	Hi-Z ²		Keep		Hi-Z	Keep
	USB_OVRCURx-DS/ USB_VBUS	Hi-Z	Hi-Z ²		Keep ³	Keep	Hi-Z	Keep
	USB_DP/USB_DM	Hi-Z	Keep-O ⁴		Keep ³	Keep	Hi-Z	Keep
RTC	RTCICx	Hi-Z	Hi-Z ²		Keep ³		Hi-Z	Keep
	RTCOU _T	Hi-Z	RTCOU _T output		Keep		Hi-Z	Keep
ACMPHS	VCOUT	Hi-Z	VCOUT output		Keep		Hi-Z	Keep
CLKOUT	CLKOUT	Hi-Z	CLKOUT output		Keep		Hi-Z	Keep
DAC	DAn	Hi-Z	D/A output retained		Keep		Hi-Z	Keep
External bus (CS, SDRAM area)	EBCLK/SDCLK	Hi-Z	High-level output		Keep		Hi-Z	Keep
	Dxx/DQxx	Hi-Z	Hi-Z		Hi-Z		Hi-Z	
	Axx/DQMx	Hi-Z	Hi-Z	Keep-O	Keep		Hi-Z	Keep
	BCx/CSx/RD/WRx/WE	Hi-Z	Hi-Z	High-level output	Keep		Hi-Z	Keep
	ALE	Hi-Z	Hi-Z	Low-level output	Keep		Hi-Z	Keep
P400/P401	CKE/SDCS/RAS/CAS	Hi-Z	Hi-Z	SDSELF.SFEN = 0: High-level output SDSELF.SFEN = 1: Low-level output	Keep		Hi-Z	Keep
	Other than function IRQ5-DS	Hi-Z	Keep-O ²		Hi-Z		Hi-Z	
Others	—	Hi-Z	Keep-O		Keep		Hi-Z	Keep

Note: Hi-Z: High-impedance

Keep-O: Output pins retain their previous values. Input pins go to high-impedance.

Keep-I: Pin states are retained the same as during periods in Normal mode.

Keep: Pin states are retained the same as during periods in Software Standby mode.

Note 1. Retains the I/O port state until the DPSBYCR.IOKEEP bit is cleared to 0.

Note 2. Input is enabled if the pin is specified as the Software Standby canceling source while it is used as an external interrupt pin.

Note 3. Input is enabled if the pin is specified as the Deep Software Standby canceling source.

Note 4. Input is enabled while the pin is used as an input pin.